

Learning Resource On Speech and Video Processing

**Module-2: Feature Extraction and Speech
Recognition**

**Chapter-5: Speech Feature Extraction and
Cepstral Analysis**

**Prepared By:
Asst. Professor
SCE, KIIT, DU, Bhubaneswar-24**

Chapter Outcomes

- After completing this module, students will be able to:
 - Describe the significance of features in signal.
 - Explain the need for feature extraction in speech processing.
 - List out different types of features.
 - Describe the concept cepstral analysis and the real and complex cepstrum.
 - Explain the basic steps involved in extracting MFCC features.
 - Describe the concept of LPCC and its relationship with LPC.
 - Differentiate between MFCC and LPCC as speech features.

Features

- **Feature** is a **numeric or categorical** representation of raw data that **isolates** a specific, meaningful attribute for analysis.
- Features may be considered as the **smallest** and **fundamental** building blocks of speech sounds.
- Features describe the **psychological** and **physiological** components that make up a single sound (phoneme).
- A **speech feature** is a measurable data point extracted from a raw audio signal.
- Features can be of various types based on different aspects;

Feature Extraction in Speech Processing

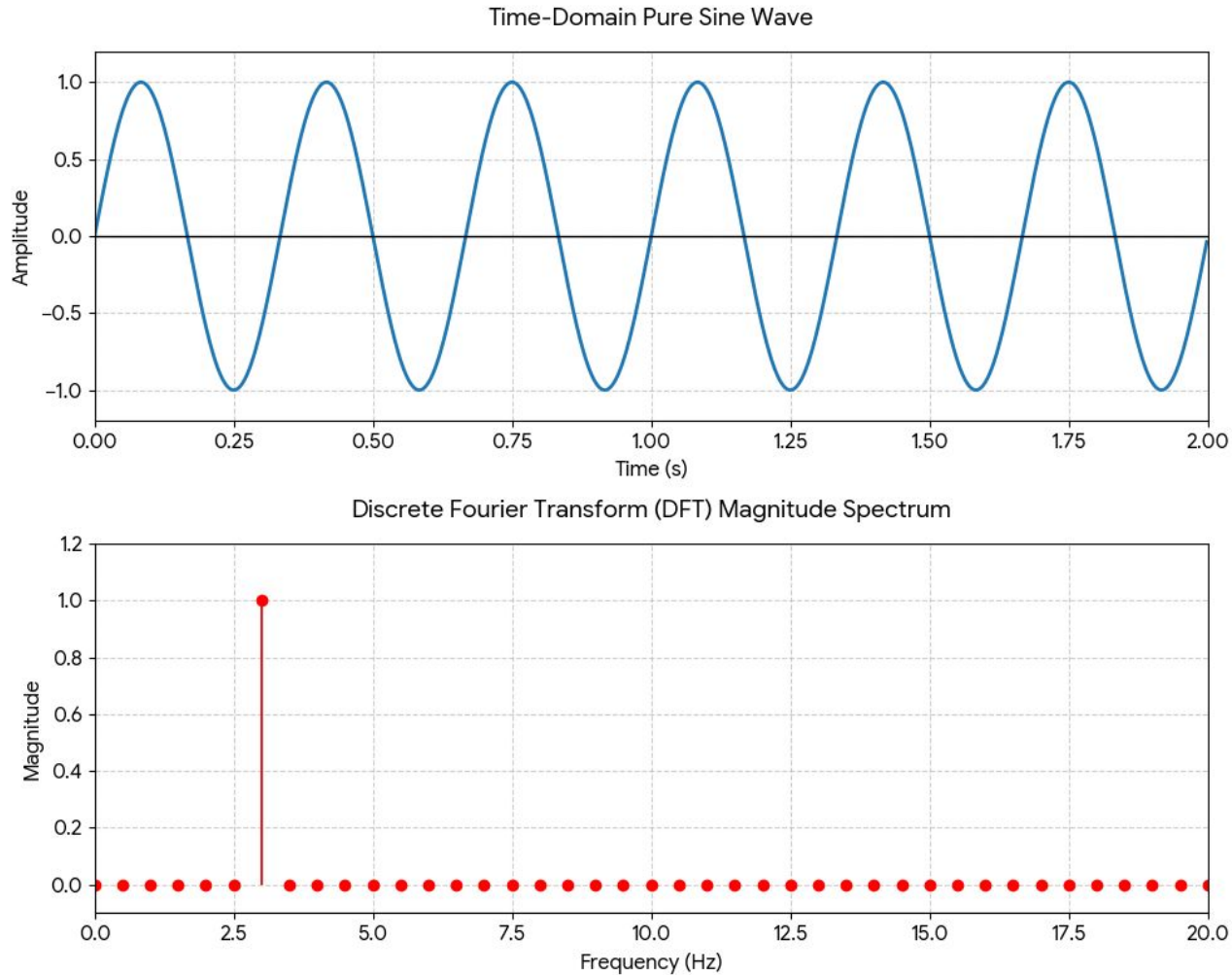
- **Feature Extraction** is the process of converting a **speech signal** into a small set of meaningful **numerical values** that represent **important characteristics** of the speech.
- A speech waveform contains a **large number of samples**. Instead of giving all these samples directly to a speech processing system, we **extract** useful information from them.
 - The **extracted features** might include information related to: Frequency content, Energy, Pitch, Spectral shape, Vocal-tract characteristics.
- Common speech features include **MFCC, LPCC, pitch, energy, and zero-crossing rate**.

Need for Feature Extraction

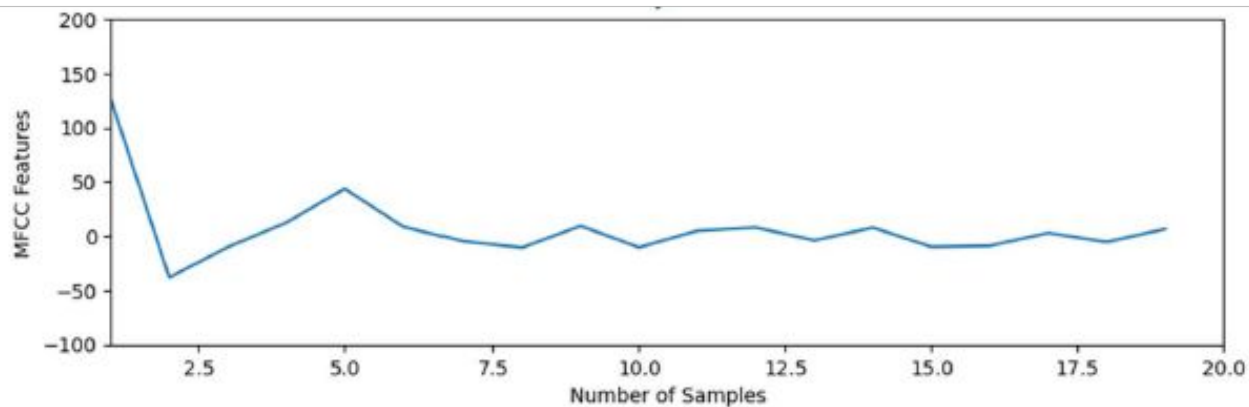
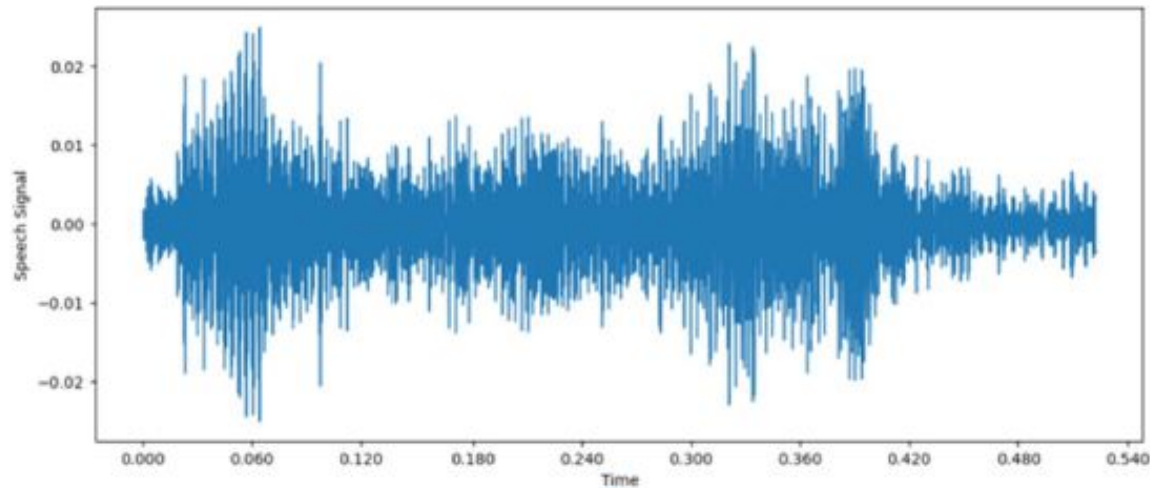
- In speech processing, feature extraction is necessary due to following reasons:
 - **High computational cost:** Speech contains a large amount of data represented in thousands of samples. Processing these samples entirely will be computationally expensive.
 - Feature extraction converts this large amount of data into a compact representation.
 - **Removing irrelevant samples:** The raw waveform contains many samples, but the speech recognition system is mainly interested in characteristics that help distinguish different sounds.
 - Thus, feature extraction keeps useful information while reducing unnecessary data.

- **Easier speech comparison:** Speech contains many types of sounds that differ in their properties.
 - Feature extraction helps us to provide a more useful representation for comparing these speech sounds.
- **Helps speech recognition systems:** Feature extraction is an important stage in applications such as speech recognition, speaker identification, emotion recognition, speech classification, voice-controlled systems.
- **Speech features** are considered as **good** if they:
 - contain relevant speech information
 - are compact
 - are robust to noise
 - are less sensitive to variation between speakers
 - help distinguish between speech sounds.

- Feature extraction **using DFT** from a sine wave



- Feature extraction **using MFCC** from a speech signal.



Types of Features

- **Static Features:** Characteristics extracted from a segment of data that describe its overall properties over a fixed window of time, ignoring how the signal changes moment to moment.
- Different types of static features are
 - **Acoustic Features:** Pitch, Energy, and Duration.
 - **Prosodic / Long-Term Features:** Rhythm, intonation, and stress patterns (utterance-level).
 - **Temporal (Time-Domain) Features:** Root Mean Square (RMS) Energy, Zero-Crossing Rate
 - **Spectral (Frequency-Domain) Features:** Spectrogram, MFCCs
 - **Glottal Features:** Jitter, Harmonics-to-Noise Ratio
 - **Deep Learning & Learned Embeddings:** Convolution Neural Network (CNN).

Types of Static Features

- **Acoustic Features:** Physical properties of the speech such as Pitch (frequency), Energy (loudness), and Duration.
 - It is divided in to various types such as temporal, spectral, cepstral and prosodic features.
- **Prosodic Features:** Spans across multiple phone segments, syllables, or entire sentences. They carry emotional and stylistic context.
 - **Fundamental Frequency (F_0 / Pitch):** The rate of vocal cord vibration; reflects intonation and stress.
 - **Duration / Rhythm:** Speaking rate, pause durations, and vowel elongation.
 - **Intensity:** Variations in volume over long segments.

Used in Speech Emotion Recognition (SER), emphasis detection, and speech synthesis (TTS).

- **Temporal (Time-Domain) Features:** Metrics extracted directly from the raw waveform over time.
 - **Root Mean Square (RMS) Energy:** Measures amplitude/loudness variations across frames.
 - **Zero-Crossing Rate (ZCR):** Tracks how often a signal changes sign (+ to -).

It is used for distinguishing voiced speech (low ZCR, high energy) from unvoiced speech / silence (high ZCR, low energy).

- **Spectral (Frequency-Domain) Features:** Features derived by transforming the signal into the frequency domain using Short-Time Fourier Transform (STFT).
 - **Spectrogram:** A visual map plotting of frequency over time, mapping signal intensity.

- **Spectral Centroid:** Pinpoints the "center of mass" of the spectrum; correlates to sound "brightness".
- **Spectral Bandwidth:** Measures the frequency spread around the centroid.
- **Cepstral Features:** Features warped mathematically to mimic how the human ear actually filters and perceives sound scales.
 - **Mel-Frequency Cepstral Coefficients (MFCCs)** are specialized features that mimic how human ears process frequencies that are used for speech recognition software.
 - **Linear Prediction Cepstral Coefficients (LPCCs)** Models the vocal tract as a digital filter, representing the resonant frequencies (formants) of speech.
 - Vocal tract filter estimation.

- **Glottal Features:** Measure the state, tension, and structural health of the speaker's vocal folds (glottis).
 - **Jitter:** Short-term variations in the pitch (F_0) period (frequency instability).
 - **Shimmer:** Short-term variations in the peak-to-peak amplitude (loudness instability).
 - **Harmonics-to-Noise Ratio (HNR):** Ratio of periodic energy to acoustic noise.

Used in Clinical speech pathology (detecting voice disorders), age/gender classification, and speaker identification.

- **Deep Learning Features:** Speech recognition uses deep learning features. Some of the features are spectrogram representations (2D spectrograms as images in CNN); Dense vector representations extracted from massive pre-trained models (wav2vec 2.0, HuBERT)

Dynamic Features

- **Dynamic Features:** They capture how the acoustic properties of a voice change over time.
 - Dynamic features track the movement, speed, and acceleration of the speech articulators (tongue, lips, jaw) across successive frames.
 - First and second derivatives of features to track how the sound changes over time.
- **Types of dynamic features**
 - **Delta:** First-order temporal derivatives of the static features.
 - Measure the rate of change of the spectrum from one frame to the next.
 - **Delta-Delta:** Second-order temporal derivatives, calculated by taking the derivative of the Delta features₁₄

Cepstral Analysis

- **Cepstrum Analysis** is a nonlinear signal processing technique.
- The cepstrum technique performed on a time-domain signal to look at periodic content in data, such as **harmonics, sidebands, and orders**.
- The word **cepstrum** was coined by **reversing the first syllable** in the word **spectrum**.
- The cepstrum exists in a domain referred to as **quefrency** (**reversal of the first syllable** in **frequency**) which has units of time.
- It is calculated by first taking the **Fourier transform** of a time signal to create a frequency spectrum; then converting it to a **log scale**; and then performing an **inverse Fourier transform** on the spectrum.

- The **cepstrum** is referred to as a “**spectrum of a spectrum**” because it involves calculating the spectrum of the time signal and then performing another spectral operation.
- In **cepstral analysis**, some terms are given new names to indicate that we are working in the cepstral domain.
 - **Quefrency**: Replaces frequency and represents the horizontal axis of the cepstrum.
 - **Saphe**: Refers to phase in the cepstral domain.
 - **Liftering**: Refers to filtering performed on cepstral coefficients.
 - **Rahmonics**: Refers to harmonic patterns observed in the cepstrum.

- In **cepstral analysis**, we first find the **magnitude spectrum** of the speech signal following which take the **log of the magnitude spectrum** and then apply the **inverse Fourier transform**.
- The result is not the original speech signal; it is a new representation called the **cepstrum**.
- The **cepstrum** helps identify **periodic patterns** in the frequency spectrum.
- **Speech harmonics** are usually **regularly spaced** in frequency, making the cepstrum useful for detecting them.
- A **prominent cepstral peak** can indicate the **pitch period** of voiced speech.
- For example, a **100 Hz** fundamental frequency corresponds to a **10 ms pitch period**, which may appear as a peak around **10 ms quefrequency**.

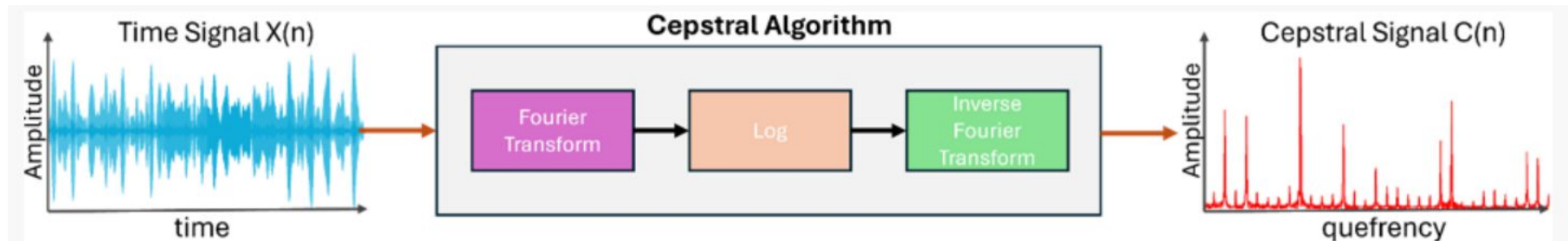
Steps to determine Cepstrum

- Steps to transform a time signal into the cepstral domain:
 - **STEP 1:** Take the time signal and **apply a FFT** on the data to translate the **time data** into **frequency data**.
 - **STEP 2:** Discard the **phase portion** of the FFT by converting the amplitude to a **$\log_{10}$ of the data**.
 - This step reduces the dynamic range of the data by compressing it to a logarithm and makes it easier to find harmonic components.
 - **STEP 3: Convert** the **frequency** signal to the **quefrequency** domain using an **Inverse Fourier Transform**. The phasing portion of an Inverse Fourier Transform rotates in the opposite direction of a Fourier Transform.
 - Since only the amplitude was kept and the log of the amplitude was applied, it converts to the quefrequency domain

Cepstral Analysis: Mathematical Representation

The mathematical representation of cepstral analysis

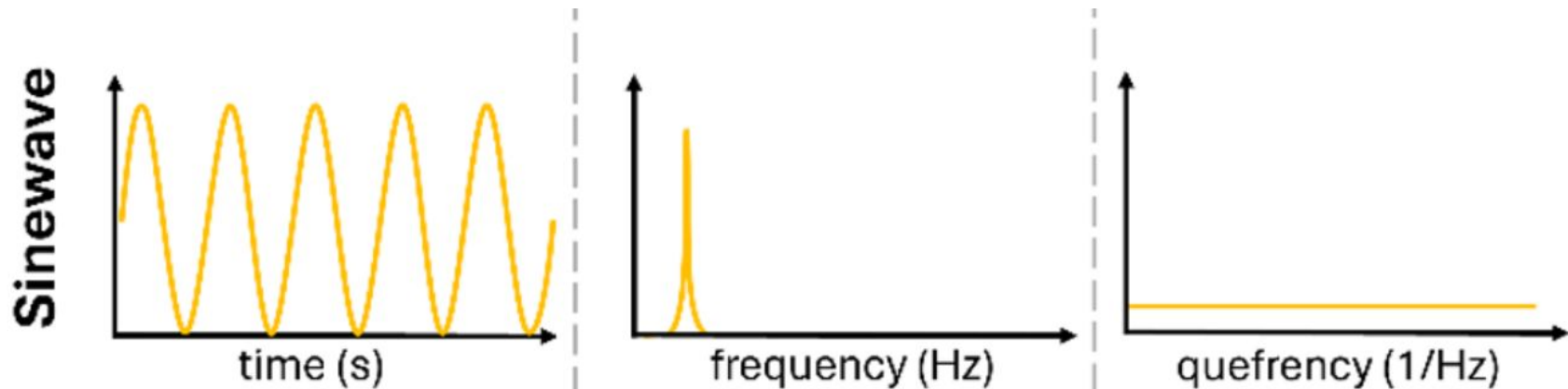
$$C[n] = \sum_{k=0}^{N-1} \log \left(\left| \sum_{n=0}^{N-1} X[n] e^{-j\left(\frac{2\pi}{N}\right)kn} \right| \right) e^{j\left(\frac{2\pi}{N}\right)kn}$$



Principle of Cepstral Analysis

- When two signals, $G(x)$ and $F(x)$, are **convolved in the time domain**, their **spectra are multiplied** in the **frequency domain**.
- In the **cepstral** domain, the **logarithm** changes **multiplication** into **addition i.e., $(G(x) + F(x))$**
- Therefore, the **cepstrum** can help us **separate** different components of a signal more easily.
- **For example**, in speech processing, it can help identify the **human voice** from **background sounds or music**.
 - A simple example is karaoke, where signal-processing techniques can be used to reduce or separate the vocal and musical components.
- This idea is also useful in applications such as speech

- Representation of sine wave in different domain:



- Two components of cepstral analysis:
 - 1.Real Cepstrum
 - 2.Complex Cepstrum

Real Cepstrum

- The **real cepstrum** of a signal $x(n)$, sometimes called simply the **cepstrum**.
 - It is calculated by determining the **natural logarithm** of **magnitude** of the **Fourier transform** of $x(n)$, then
 - obtaining the **inverse Fourier transform** of the resulting sequence.
- **Real cepstrum** of a digital signal $x(n)$ is defined as:

$$c(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln |X(\omega)| e^{j\omega n} d\omega$$

Where, n is quefrency index; $X(\omega)$ is the frequency spectrum; ω is the digital angular frequency; $e^{j\omega n}$ is the complex exponential

- The Real Cepstrum uses only the **magnitude** of the signal's

Steps to calculate Real Cepstrum

- **Steps to calculate Real Cepstrum: ($C_r(n)$)**
 - Compute the DFT (F) of the signal $x(n)$.
 - Take the absolute magnitude of the result.
 - Compute the natural logarithm of that magnitude.
 - Take the Inverse DFT (IDFT)

$$c_r(n) = \mathcal{F}^{-1} \{ \ln |\mathcal{F}\{x(n)\}| \}$$

- Because the **real cepstrum discards phase data**, it is used when you only need to **analyze the periodicities** or the **overall shape (envelope)** of a signal.

Application of Real Cepstrum

- Applications of Real Cepstrum:
 - **Speech Analysis:**
 - It extracts the pitch (fundamental frequency) of a speaker.
 - It also helps isolate the spectral envelope, which is used in speech recognition to identify vowels and consonants.
 - **Audio Processing:**
 - It is the foundation for creating Mel-Frequency Cepstral Coefficients (MFCCs), the core feature used by voice assistants (like Alexa or Siri) to understand commands.

Complex Cepstrum

- The **complex cepstrum** of a sequence $x(n)$ is calculated by finding the complex natural logarithm of the Fourier transform of $x(n)$, then the inverse Fourier transform of the resulting sequence.
- The **complex cepstrum** is defined as:

$$\hat{x}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln X(\omega) e^{j\omega n} d\omega$$

- The Complex Cepstrum uses both the **magnitude and the phase**, making it a **fully reversible transformation**.

Steps to calculate Complex Cepstrum

- **Steps to calculate Complex Cepstrum: (Cc(n))**

- Compute the DFT (F) of the signal x(n).
- Express the result in polar form.

$$X(\omega) = |X(\omega)|e^{j\angle X(\omega)}$$

- Compute the complex logarithm.

$$\ln(X(\omega)) = \ln|X(\omega)| + j\angle X(\omega)$$

- Take the Inverse DFT (IDFT)

$$c_c(n) = \mathcal{F}^{-1} \{ \ln(\mathcal{F}\{x(n)\}) \}$$

- Because the **complex cepstrum** preserves **phase data**, it is **used** when you need to **modify the signal** and **convert**

Applications of Complex Cepstrum

- **Echo Removal & Dereverberation:** It identifies the exact delay time of an echo.
 - By removing the echo peak in the "quefreny" domain and converting it back, it completely removes echoes from audio or telecom recordings.
- **Speech Synthesis (Text-to-Speech):** It allows engineers to separate the vocal tract filter from the glottis source.
 - Designers can then modify the pitch or duration independently to create natural-sounding robotic speech.
- **Deconvolution of Blurring:** It restores signals or one-dimensional imaging data that have been distorted or blurred by a transmission channel.

Cepstral Analysis for Speech signal

- **Cepstral Analysis** is primarily used in **speech processing** to separate the **excitation source (pitch)** from the **vocal tract filter (formants)**.
- This mathematical process **deconvolves** a speech signal by transforming a **multiplicative relationship** in the **frequency domain** into an **additive one in the cepstral domain**.
- Speech $s(n)$ is generated by **convolving** an **excitation source $e(n)$** (glottal pulses or air noise) with a **vocal tract filter $h(n)$** :

$$s(n) = e(n) * h(n)$$

- In the **standard frequency domain**, this **convolution** becomes a **multiplication**.

- Linear filters **cannot easily separate** multiplied signals.
- **Logarithm transforms** takes the mathematical relationship from **multiplicative to additive**:

$$\log |S(w)| = \log |E(w)| + \log |H(w)|$$

where $\log |H(w)|$ = Spectral envelope (vocal tract shape)

$\log |E(w)|$ = rapidly varying ripples (harmonics/pitch).

- Because the **vocal tract** changes **slowly**, its **log-spectrum** is **smooth (low frequency)**.
- Because the **glottal excitation** consists of **periodic pulses**, its **log-spectrum** has **sharp, rapid ripples (high frequency)**.

Real Cepstrum of a speech signal

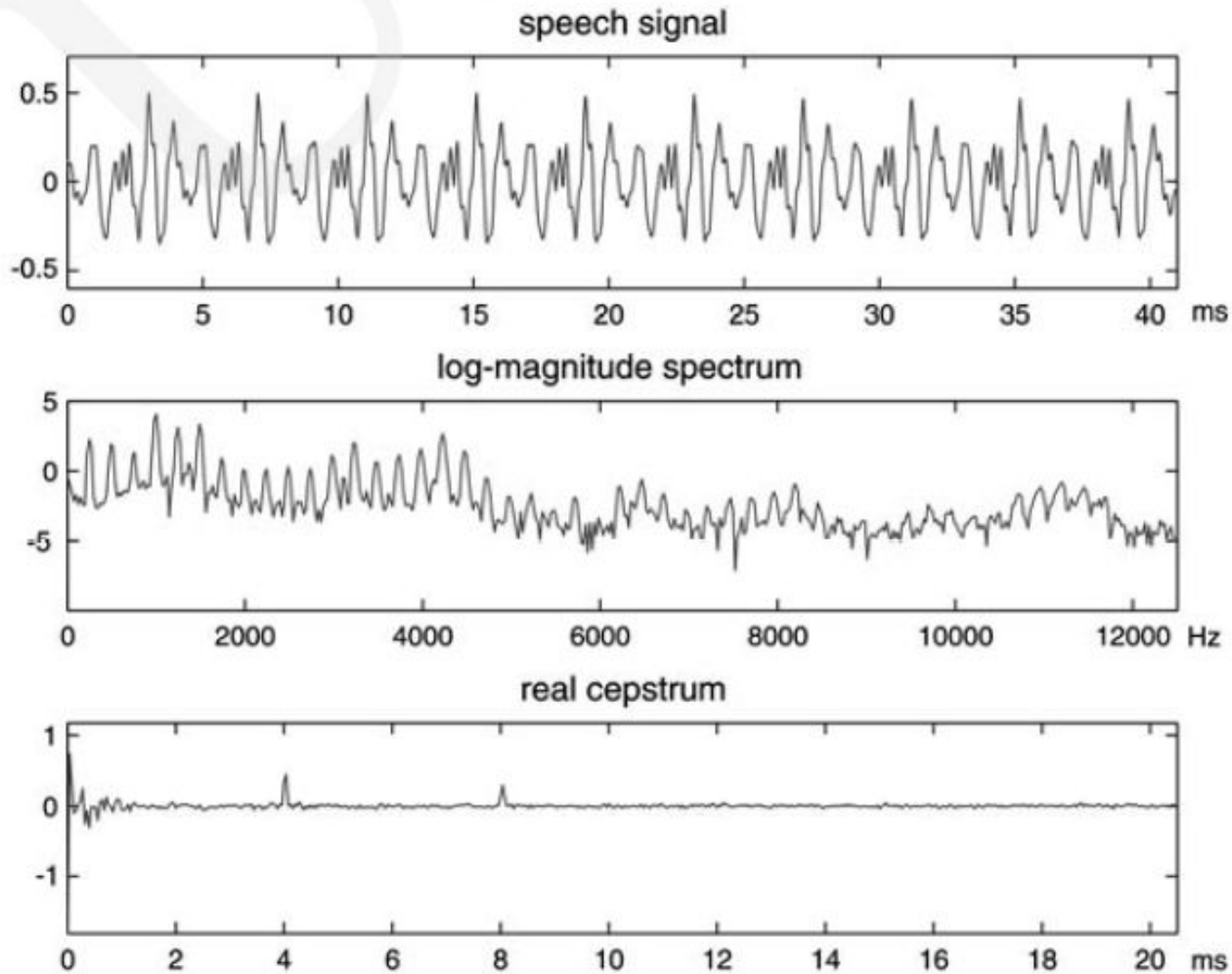
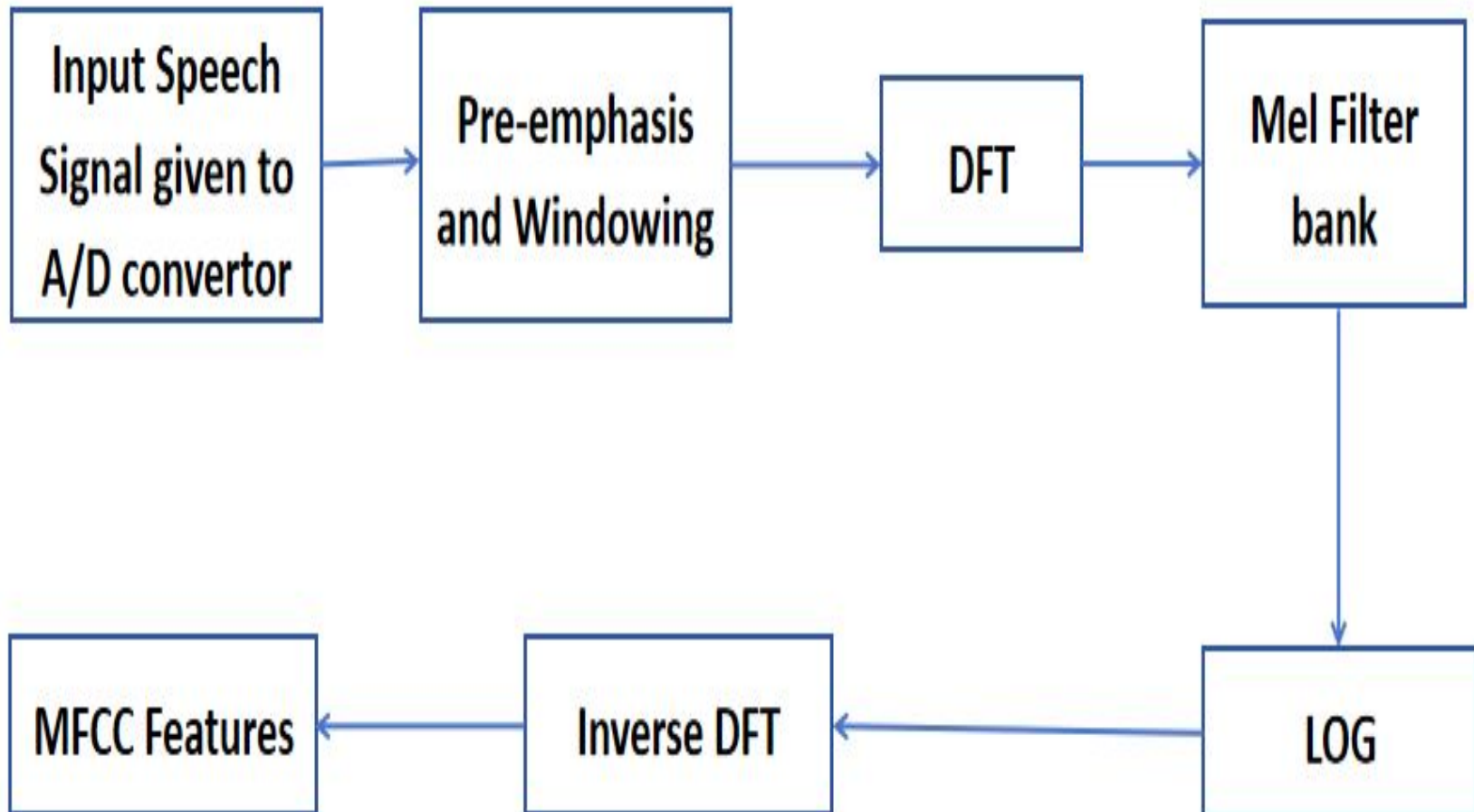


Fig. 1. A segment of speech signal, log-magnitude spectrum and cepstrum

Mel-frequency Cepstral Coefficients

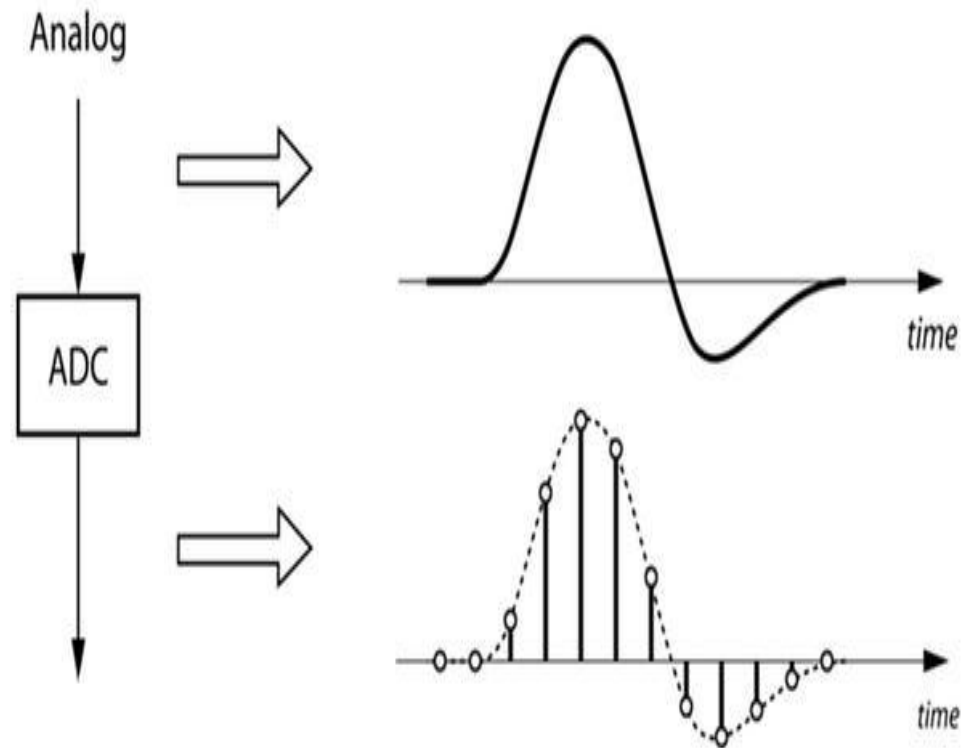
- Humans do not perceive pitch linearly. Humans are much better at detecting small changes in low frequencies than in high frequencies.
- **Mel-Frequency Cepstral Coefficients (MFCCs)**
 - Most widely used feature extraction technique in traditional speech processing.
 - They are used to represent the power spectrum of an audio signal based on how the human ear actually perceives sound.
 - MFCC provides a compact matrix (typically 12–13 coefficients per time frame) that captures the fingerprint of the sound.

Steps to extract MFCCs



A/D Conversion

1. A/D Conversion: samples the audio clips and digitizes the content, i.e. converting the analog signal into discrete space. A sampling frequency of 8 or 16 kHz is often used.

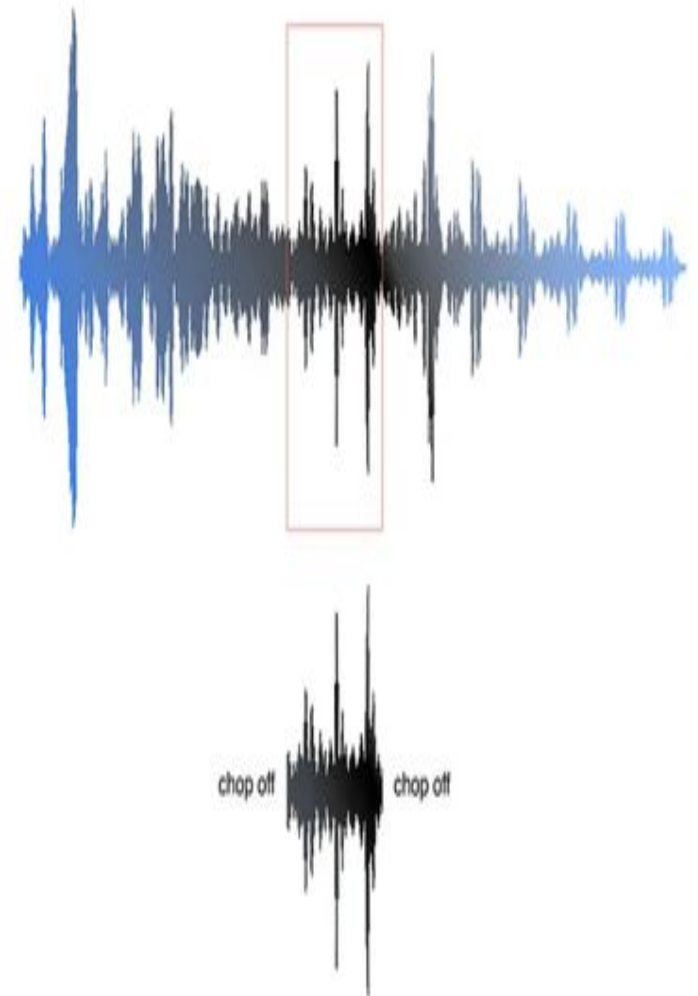
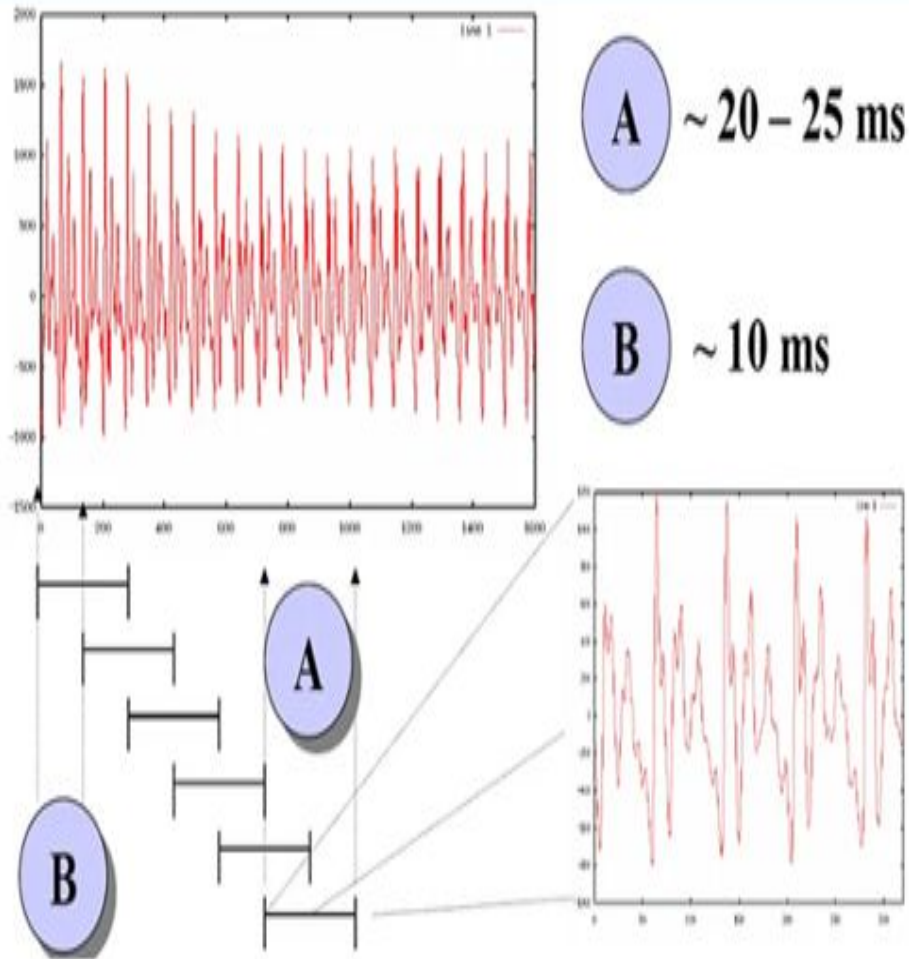


2. Pre-emphasis: boosts the amount of energy in the high frequencies.

- For voiced segments like vowels, there is more energy at the lower frequencies than the higher frequencies.
- This is called spectral tilt which is related to the glottal source (how vocal folds produce sound).
- Boosting the high-frequency energy makes information in higher formants more available to the acoustic model.
- Pre-emphasis is used to make the system less susceptible to noise introduced in the process later.

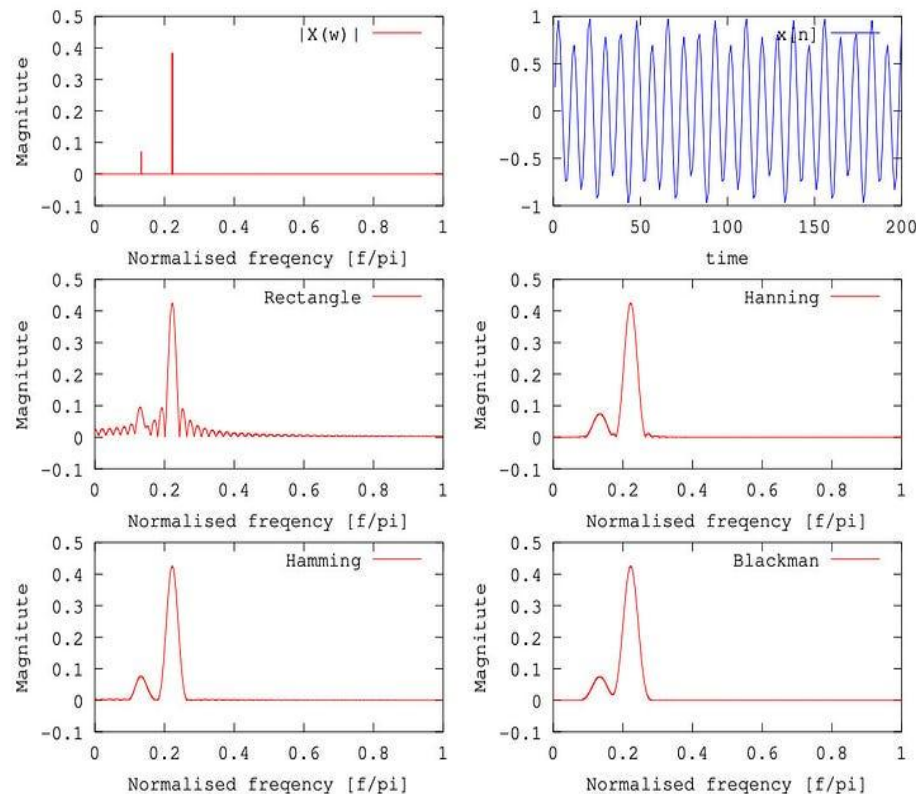
3. Windowing :

- Windowing involves the **slicing** of the audio waveform into sliding frames. It cannot be just chop it off at the edge of the frame.
 - The suddenly fallen in amplitude will create a lot of noise that shows up in the high-frequency.
- To slice the audio, the amplitude should **gradually drop off** near the edge of a frame.
- It can be written as $\mathbf{x}[\mathbf{n}] = \mathbf{w}[\mathbf{n}] \mathbf{s}[\mathbf{n}]$; where, $\mathbf{w}$ is the window applied to the original audio clip in the time domain.
- w may be the **Hamming** window and the **Hanning** window.



- Speech is **not stationary** for a long time, but it can be assumed approximately **stationary over a short interval**.
- The speech signal is divided into **small frames/windows**:
 - $A \approx 20\text{--}25$ ms: **frame length** used for speech analysis.
 - $B \approx 10$ ms: **frame shift** (the window moves by about 10 ms).
- Because the frame length is **larger** than the frame shift, consecutive **frames overlap**.
- The highlighted portion on the right shows one short speech segment being selected for analysis.
- The “**chop off**” labels indicate that only the selected short segment is analyzed at a time, rather than the entire speech signal.

- The chopped frame with **Hamming** and **Hanning** maintains the original frequency information better with less noise compared to a rectangle window.



- When we analyze only a **finite** portion of the signal, we effectively apply a window. This spreads the spectral energy around the actual frequency, producing spectral leakage.
- **Effect of different windows**
 - **Rectangle:** Narrow main peak but large side lobes; more spectral leakage.
 - **Hanning (Hann):** Reduces side lobe; less leakage, but slightly wider main lobe.
 - **Hamming:** Good reduction of side lobes while maintaining reasonable frequency resolution.
 - **Blackman:** Strongest side-lobe suppression among these, but has the widest main lobe.

4. Discrete Fourier Transform (DFT): Apply DFT to extract information in the frequency domain.

5. Mel filterbank

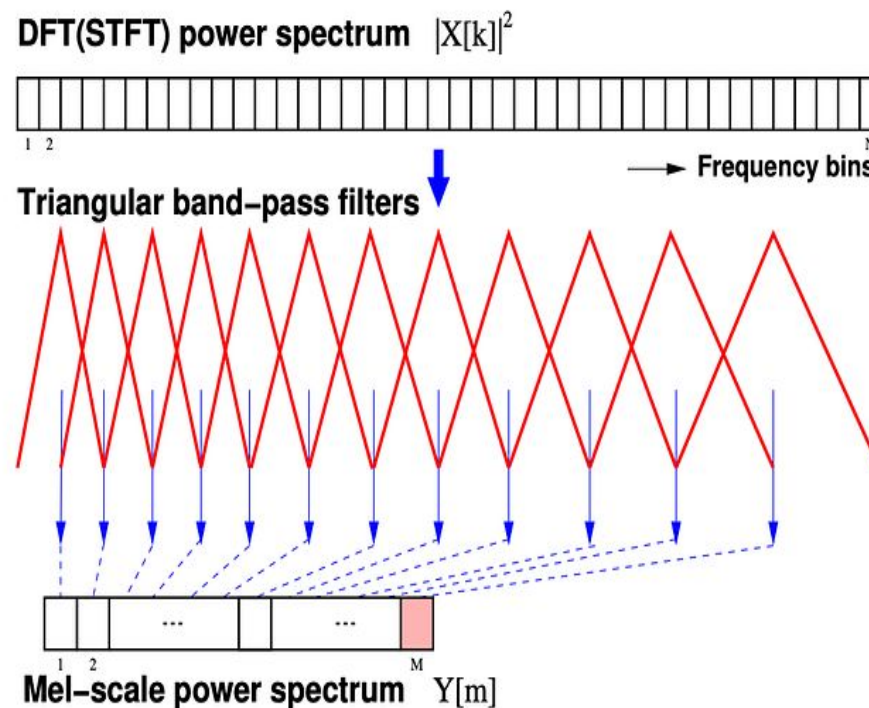
- The equipment measurements are not the same as our hearing perception.
- For humans, the perceived loudness changes according to frequency.
- Also, perceived frequency resolution decreases as frequency increases i.e. humans are less sensitive to higher frequencies.
- The **Mel scale** maps the measured frequency to that we perceived in the context of frequency resolution. All these mappings are non-linear.
- Apply **triangular band-pass filters** to convert the frequency information to mimic what a human perceived⁴⁰

- **Square the output of the DFT** that reflects the power of the speech at each frequency ($x[k]^2$) and it is the **DFT power spectrum**.
- Triangular Mel-scale filter banks transform it to **Mel-scale power spectrum**.
- The output for each Mel-scale power spectrum slot represents the energy from a number of frequency bands that it covers. This mapping is called the **Mel Binning**.
- The precise equations for slot m will be:

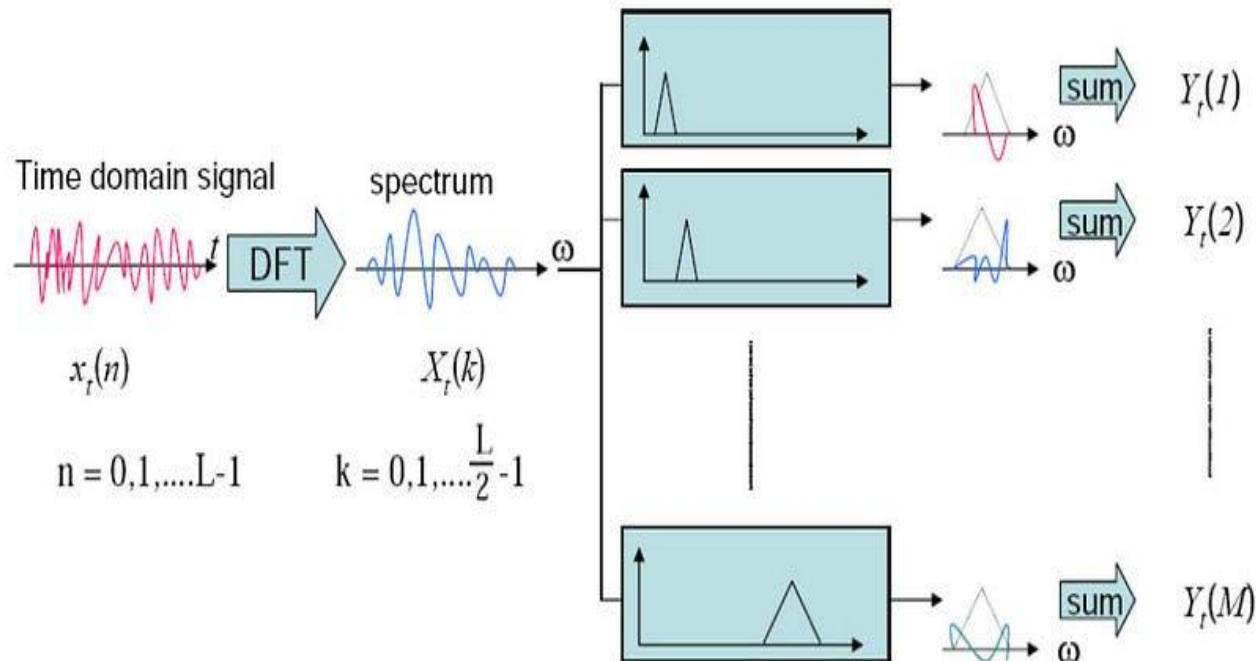
$$Y_t[m] = \sum_{k=1}^N W_m[k] |X_t[k]|^2$$

where k : DFT bin number ($1, \dots, N$)
 m : mel-filter bank number ($1, \dots, M$)

- The Triangular bandpass is wider at the higher frequencies to reflect human hearing is less sensitivity in high frequency.
- All these efforts try to mimic how the **basilar membrane in ear senses the vibration of sounds.**



- The basilar membrane has about 15,000 hairs inside the cochlear at birth.
- The frequency response of the hairs is approximated by triangles in Mel filterbank.
- It imitate how ears perceive sound through those hairs.



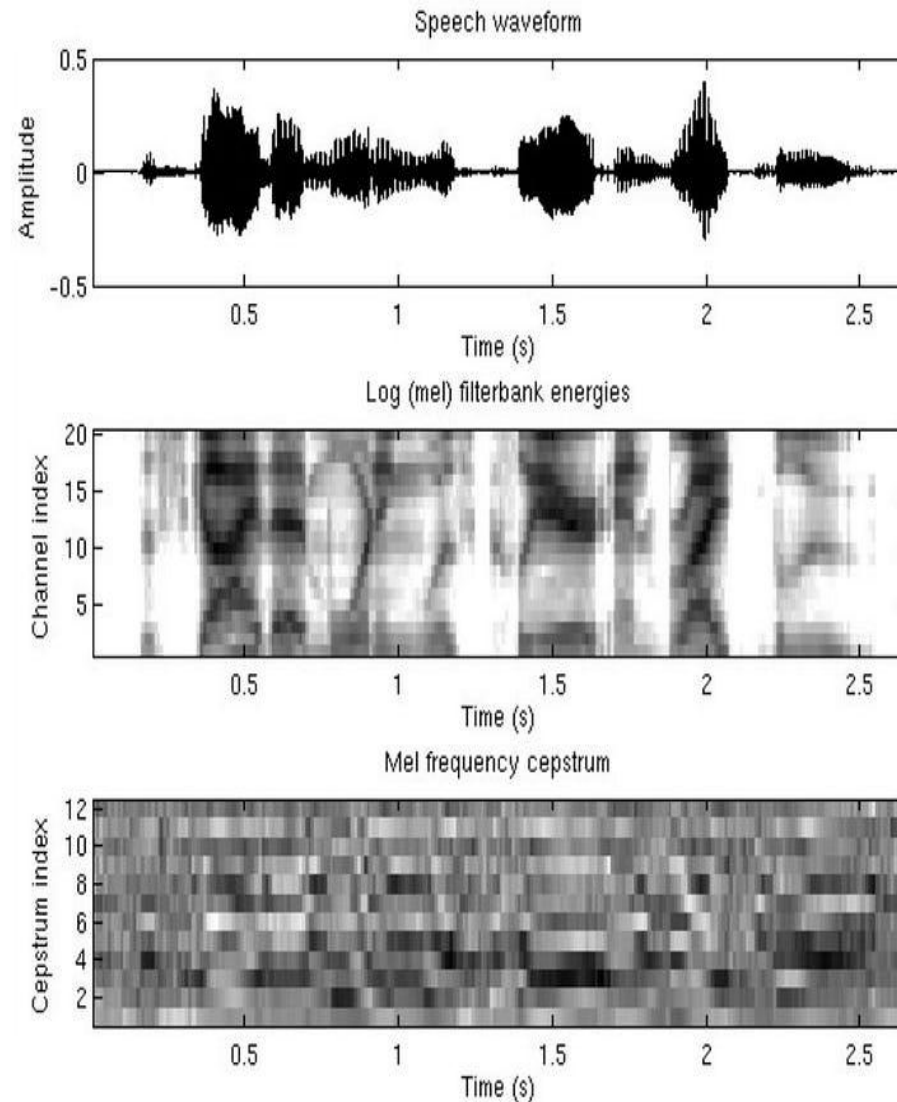
6. Log:

- Mel filterbank outputs a **power spectrum**.
- Humans are less sensitive to small energy change at high energy than small changes at a low energy level.
- It is logarithmic.
- Take the **log out** of the **output of the Mel filterbank**.
- It reduces the acoustic variants that are not significant for speech recognition.

7. IDFT:

- Inverse Discrete Fourier Transformation is taken to separate the pitch information from the formants.
- MFCC just takes the first 12 cepstral values and discard others.

- Visualization of the 12 Cepstrum coefficients.



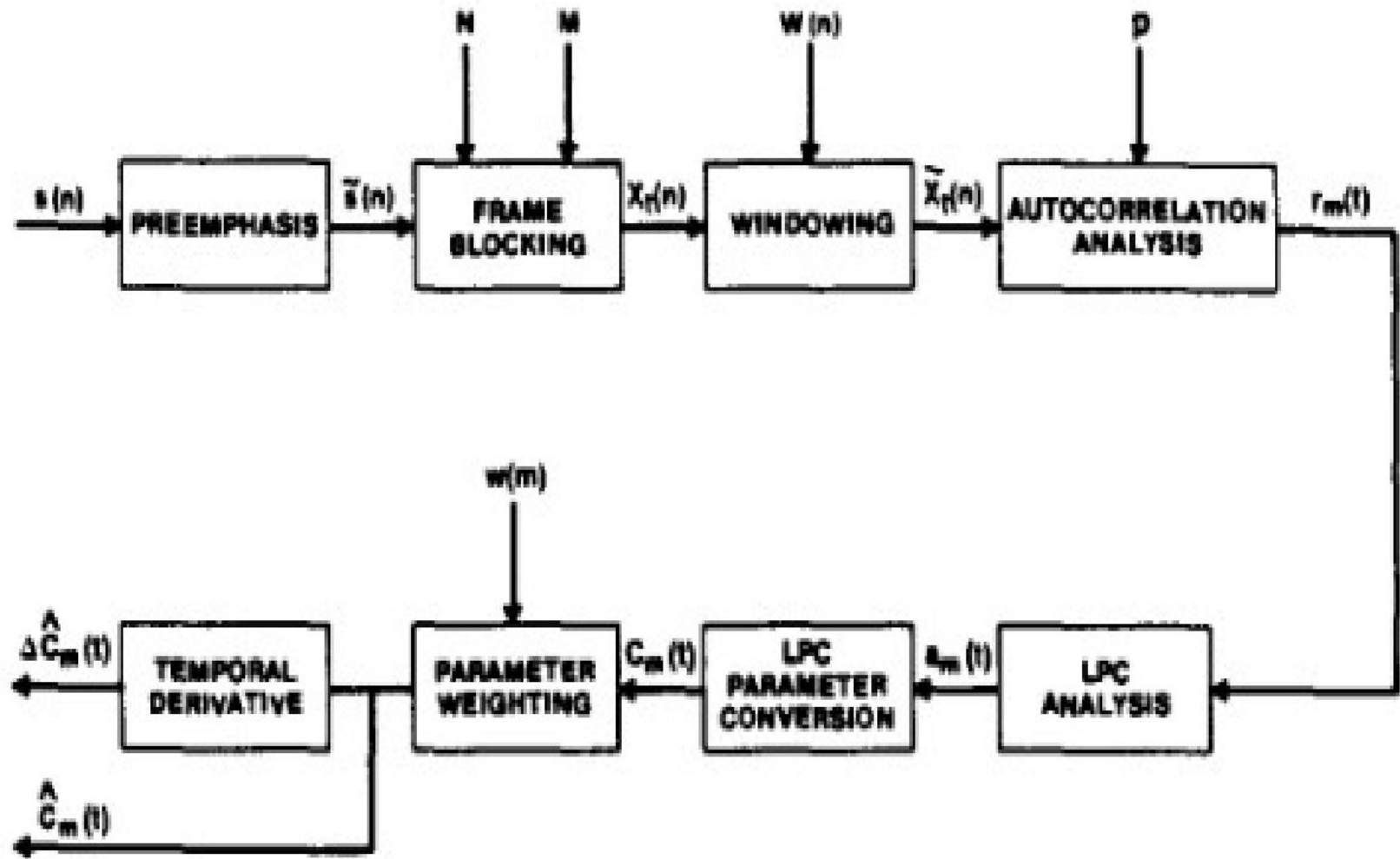
Applications of MFCC

- **Automated Speech Recognition (ASR):** Converts spoken language into text (used in virtual assistants like Siri, Alexa).
- **Phoneme Classification:** Identifies individual structural sounds (like "sh" or "ah") to map out spoken words.
- **Keyword Spotting:** Powers voice-activation triggers like "Hey Siri" or "OK Google" using low computational footprint pipelines.
- **Speaker Identification:** Recognizes who is speaking out of a known database by capturing the unique physical structure of their vocal tract.
- **Speaker Verification:** Acts as a security layer to verify a user's voiceprint passkey for banking or secure device access.

Linear Prediction Cepstral Coefficients (LPCC)

- **LPCC (Linear Prediction Cepstral Coefficients)** is a digital feature extraction technique used in speech processing.
- It calculates the **smooth spectral envelope** of a vocal signal by merging **Linear Predictive Coding (LPC)** with **Cepstral Analysis**.
- While MFCC models human-hearing, LPCC mathematically models the biological mechanism of the **human vocal tract** (throat, tongue, and mouth).
- It isolates the **source filter (vocal cords)** from the **vocal tract filter (throat, tongue, mouth)**.

Steps to extract LPCC



- **Pre-emphasis:** The digital speech signal $s(n)$ passes through a **first order FIR filter** to balance the spectrum.

$$H(z) = 1 - \alpha z^{-1}, \quad 0.9 \leq \alpha \leq 1.0.$$

- It amplifying higher frequencies that naturally drop off during human speech production.
- The output of the preemphasis, $s(n)'$ of the signal $s(n)$, where $\alpha \approx 0.95$ to 0.98 , is

$$s(n)' = s(n) - \alpha s(n-1)$$

Frame blocking and Windowing

- **Frame blocking** is segmenting the speech waveform into **short overlapping frames** so that the signal can be treated as approximately **stationary**.
 - The signal $s(n)$ is blocked into N number of samples with adjacent frames being separated by M samples.
 - If there are L frames in speech signal and **L th frame** is denoted by $x_l(n)$.
- **Windowing:** The continuous audio is sliced into short, overlapping frames (typically 20–30 milliseconds).
 - A window function (Hamming Window) is multiplied by each frame.
 - It eliminates edge discontinuities and minimizes spectral leakage.

- **Autocorrelation analysis:** Each windowed frame undergoes autocorrelation matching to find **periodic repetitions** and **minimize errors** between actual and predicted signal frames.
 - It compute $p+1$ autocorrelation values for each frame.
 - P is 8 to 16 generally.
 - The zeroth value of the auto correlation $R_l(0)$ gives l^{th} frame energy.
 - Higher lags describe short-term similarity between samples.

These values become the input to the **LPC coefficient computation.**

- **LPC analysis:** The LPC analysis converts $p+1$ autocorrelation values into an **LPC parameter set**.
- Set may be LPC coefficient, reflection coefficient, log area ratio coefficients and cepstral coefficients.
- Method of converting autocorrelation coefficients to LPC parameter is called Durbin method.
- Using the Levinson-Durbin recursion algorithm, the system extracts linear prediction coefficients.
- At each recursion, a reflection coefficient is obtained and the predictor coefficients are updated.
- The recursion also tracks the residual prediction error energy.

LPC Parameter Conversion

- **LPC parameter conversion:** LPC parameters set that can be derived directly from LPC coefficient are cepstral coefficients C_m via a direct recursive formula. The recursion used is:

$$c_0 = \ln \sigma^2$$

$$c_m = a_m + \sum_{k=1}^{m-1} \left(\frac{k}{m} \right) c_k a_{m-k}, \quad 1 \leq m \leq p$$

$$c_m = \sum_{k=1}^{m-1} \left(\frac{k}{m} \right) c_k a_{m-k}, \quad m > p,$$

- This final conversion yields highly uncorrelated robust parameters.
- $c(m)$ derived recursively from $a_1 \dots a_p$ and gain.
- The **gain term** is included in the cepstral representation is
- Q cepstral coefficients are generated from the LPC parameters, where Q may be greater than the LPC order p . ($Q \approx 3/2p$).
- **LPC Coefficients:**
 - One vector per analysis frame is produced. Each vector can contain weighted LPC-derived cepstral coefficients.
 - Temporal cepstral derivatives may be appended to capture spectral dynamics.
 - The resulting feature sequence is Q cepstral coefficient and appended vector of Q cepstral time derivative .

$$V(t) = [c_1(t), c_2(t), \dots, c_Q(t), \Delta c_1(t), \Delta c_2(t), \dots, \Delta c_Q(t)]$$

Applications of LPCC

- Because LPCC accurately maps individual physical attributes of a person's voice apparatus, it excels in tasks focusing on identity and nuance over language semantics:
1. **Speaker Identification & Verification:** Biometric security systems use it to recognize an individual's distinct voice print.
 2. **Speech Emotion Recognition (SER):** Subtle changes in throat tension due to anger, sadness, or excitement drastically affect the LPCC envelope.
 3. **Speech Compression & Coding:** Telecommunication networks use it to transmit high-quality speech over incredibly low bitrates.

Difference between MFCC and LPCC

Metric	LPCC (Linear Prediction Cepstral Coeff.)	MFCC (Mel-Frequency Cepstral Coeff.)
Underlying Model	Mimics the biological vocal tract	Mimics the psychological human ear
Frequency Scale	Linear frequency scale	Non-linear Mel frequency scale
Noise Tolerance	Vulnerable; performance drops in noisy environments	Highly robust against background noise
Best Used For	Speaker ID, biometric security, emotion detection	Automatic Speech Recognition (ASR), translation